

Proteins, Fats & Vitamins – Page 422

Source: Lucent's General Knowledge – Biology

Biology: Proteins, Fats & Vitamins

50 Questions • 30 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. On the basis of chemical composition, proteins are divided into how many types?

- A. 2
- B. 3
- C. 4
- D. 5

Ans: B

Q2. Simple Protein consists of only:

- A. Fatty acids
- B. Amino acids
- C. Nucleic acids
- D. Carbohydrates

Ans: B

Q3. Which is an example of Conjugated Protein?

- A. Albumin
- B. Histones
- C. Haemoglobin (having some other chemical compound in addition to amino acids)
- D. Globulin

Ans: C

Q4. Derived Protein is obtained from:

- A. Fats
- B. Partial hydrolysis of natural protein
- C. Carbohydrates
- D. Minerals

Ans: B

Q5. Animal protein is called first class protein because:

- A. It is cheap
- B. It contains all the essential amino acids
- C. It is vegetarian
- D. It is easy to digest

Ans: B

Q6. Egg contains how many essential amino acids?

- A. 10
- B. 15
- C. 20 (all)
- D. 5

Ans: C

Q7. Proteins are important for:

- A. Only energy
- B. Physical growth, protoplasm and formation of cells, tissues
- C. Only bones
- D. Only blood

Ans: B

Q8. Lack of proteins hampers growth. This causes a disease called: A. Rickets B. Kwashiorkor C. Scurvy D. Beriberi	Ans: B
Q9. In Kwashiorkor disease, which parts of children are affected? A. Only legs B. Stomach comes out, hands and legs get slimmed C. Only hands D. Only head	Ans: B
Q10. Marasmus is caused by: A. Excess protein B. Loosened deficiency of protein and energy C. Excess vitamins D. Iron deficiency	Ans: B
Q11. Biological catalysts which speed up chemical reactions and remain unchanged at 37°C are called: A. Hormones B. Enzymes C. Vitamins D. Minerals	Ans: B
Q12. Human nails are made up of which protein? A. Albumin B. Keratin C. Casein D. Globulin	Ans: B
Q13. Soybean is the richest source of: A. Fat B. Protein C. Carbohydrate D. Vitamin C	Ans: B
Q14. When egg is heated, the transparent liquid portion turns solid. This happens due to: A. Evaporation B. Thermal denaturation of protein C. Chemical reaction with air D. Loss of water	Ans: B

Q15. Keratin is a protein which protects: A. Bones B. Epithelial cells from damage C. Blood cells D. Nerve cells	Ans: B
Q16. Fats are esters of: A. Amino acid and glucose B. Glycerol and fatty acid C. Nucleic acid and protein D. Starch and sugar	Ans: B
Q17. Fats are of how many types? A. 3 B. 2 – Saturated and Unsaturated fatty acids C. 4 D. 1	Ans: B
Q18. Saturated fatty acids are found in: A. Fish oil and vegetable oil B. Fat (solid at 20°C temperature) C. Only soybean oil D. Only coconut water	Ans: B
Q19. Unsaturated fatty acids are found in: A. Solid fats only B. Fish oil and vegetable oil (liquid form - called oil) C. Butter only D. Cheese only	Ans: B
Q20. An adult person should get approximately what percentage of energy from fat? A. 10-15% B. 20-30% C. 40-50% D. 5-10%	Ans: B
Q21. 1 gram of fat liberates how many kilo calories of energy? A. 4.1 B. 9.3 C. 7.0 D. 5.5	Ans: B
Q22. Fat stores maximum energy per: A. Milliliter B. Gram C. Kilogram D. Liter	Ans: B

Q23. Fat provides energy to the body and: A. Increases body weight only B. Prevents the loss of heat and remains under the skin C. Causes diseases D. Has no function	Ans: B
Q24. Rancidity is the oxidation of fat. It changes the: A. Color only B. Taste and smell of food C. Volume D. Weight	Ans: B
Q25. Due to lack of fat, skin gets: A. Darker B. Dry and body weight decreases C. Thicker D. Smoother	Ans: B
Q26. Vitamin was first coined/invented by: A. Louis Pasteur B. Sir F.G. Hopkins C. Robert Koch D. Alexander Fleming	Ans: B
Q27. The term 'Vitamine' was first used by: A. Hopkins B. Funk C. Pasteur D. Darwin	Ans: B
Q28. On the basis of solubility, vitamins are of how many types? A. 3 B. 2 – Water soluble and Fat soluble C. 4 D. 1	Ans: B
Q29. Which vitamins are water soluble? A. A and D B. B and C C. E and K D. A and K	Ans: B
Q30. Which vitamins are fat soluble? A. B and C B. A, D, E and K C. Only B12 D. Only C	Ans: B

Q31. Chemical name of Vitamin A is: A. Thiamine B. Retinol C. Riboflavin D. Niacin	Ans: B
Q32. Deficiency of Vitamin A causes: A. Beriberi B. Night blindness, Xerophthalmia C. Scurvy D. Pellagra	Ans: B
Q33. Chemical name of Vitamin B1 is: A. Riboflavin B. Thiamine C. Niacin D. Pyridoxine	Ans: B
Q34. Deficiency of Vitamin B2 (Riboflavin) causes: A. Night blindness B. Cracking of skin, reddish eye, cracking at corners of lips, mouth ulcers C. Scurvy D. Rickets	Ans: B
Q35. Deficiency of Vitamin B3 (Nicotinamide/Niacin) causes: A. Rickets B. Whitening of hair, mental retardation, Pellagra C. Scurvy D. Night blindness	Ans: B
Q36. Deficiency of Vitamin B5 (Pantothenic acid) causes: A. Scurvy B. Burning sensation, Dermatitis C. Rickets D. Night blindness	Ans: B
Q37. Chemical name of Vitamin B6 is: A. Riboflavin B. Pyridoxine C. Thiamine D. Niacin	Ans: B
Q38. Deficiency of Vitamin B6 causes: A. Night blindness B. Anemia, skin disease, Depression C. Rickets D. Scurvy	Ans: B

Q39. Deficiency of Vitamin B7 (Biotin) causes: A. Scurvy B. Paralysis, body pain, hair falling, scaly and itchy skin C. Pellagra D. Rickets	Ans: B
Q40. Chemical name of Vitamin B9 is: A. Cyanocobalamin B. Folic acid C. Biotin D. Thiamine	Ans: B
Q41. Deficiency of Vitamin B9 (Folic acid) causes: A. Night blindness B. Pernicious Anemia, jaundice C. Rickets D. Pellagra	Ans: B
Q42. Chemical name of Vitamin B12 is: A. Folic acid B. Cyanocobalamin C. Pyridoxine D. Riboflavin	Ans: B
Q43. Deficiency of Vitamin B12 causes: A. Scurvy B. Anemia, dysentery C. Pellagra D. Rickets	Ans: B
Q44. Chemical name of Vitamin C is: A. Retinol B. Ascorbic acid C. Calciferol D. Tocopherol	Ans: B
Q45. Deficiency of Vitamin C causes: A. Rickets B. Scurvy, swelling of gums or bleeding gums C. Night blindness D. Beriberi	Ans: B
Q46. Sources of Vitamin A include: A. Sea fish only B. Milk, Egg, Cheese, Green vegetable, Carrot C. Only fruits D. Only grains	Ans: B

Q47. Kwashiorkor disease typically develops in children due to:

- A. Excess protein
- B. Protein deficiency while caloric intake may be adequate
- C. Vitamin D deficiency
- D. Iron deficiency

Ans: B

Q48. Proteins take part in formation of:

- A. Only bones
- B. Cells, protoplasm and tissues
- C. Only blood
- D. Only skin

Ans: B

Q49. Sources of Vitamin B1 (Thiamine) include:

- A. Fish liver oil
- B. Pulse, Liver, Egg, Milk, eggs etc.
- C. Only sea fish
- D. Only iodized salt

Ans: B

Q50. Vitamin B1 (Thiamine) deficiency causes which disease?

- A. Scurvy
- B. Beriberi
- C. Pellagra
- D. Rickets

Ans: B

Answer Key & Explanations

Q1. Answer: B — 3

Page 422: Proteins are divided into three types: Simple Protein, Conjugated Protein, and Derived Protein.

Topic: Proteins

Q2. Answer: B — Amino acids

Page 422: Simple Protein consists of only amino acids. Examples: Albumins, Globulins, Histones.

Topic: Proteins

Q3. Answer: C — Haemoglobin (having some other chemical compound in addition to amino acids)

Page 422: Conjugated Protein has some other chemical compound in addition to amino acids, e.g. Haemoglobin.

Topic: Proteins

Q4. Answer: B — Partial hydrolysis of natural protein

Page 422: Derived Protein is derived from the partial hydrolysis of natural protein. Examples: Peptone, Peptide, Proteases.

Topic: Proteins

Q5. Answer: B — It contains all the essential amino acids

Page 422: Animal protein is called first class protein because it contains all the essential amino acids.

Topic: Proteins

Q6. Answer: C — 20 (all)

Page 422: Egg contains all the essential amino acids.

Topic: Proteins

Q7. Answer: B — Physical growth, protoplasm and formation of cells, tissues

Page 422: Proteins are important for physical growth, protoplasm and formation of cells, tissues.

Topic: Proteins

Q8. Answer: B — Kwashiorkor

Page 422: Lack of proteins causes Kwashiorkor and Marasmus in children.

Topic: Proteins

Q9. Answer: B — Stomach comes out, hands and legs get slimmed

Page 422: In Kwashiorkor, the stomach comes out, and hands and legs of children get slimmed. Caused by protein deficiency.

Topic: Proteins

Q10. Answer: B — Loosened deficiency of protein and energy

Page 422: Marasmus is caused by loosened deficiency of protein and energy together.

Topic: Proteins

Q11. Answer: B — Enzymes

Page 422: Biological catalysts (enzymes) speed up chemical reactions and remain unchanged after reaction at 37°C.

Topic: Proteins

Q12. Answer: B — Keratin

Page 422: Human nails are made up of protein Keratin.

Topic: Proteins

Q13. Answer: B — Protein

Page 422: Soybean is the richest source of protein.

Topic: Proteins

Q14. Answer: B — Thermal denaturation of protein

Page 422: When egg is heated, the protein turns solid due to thermal denaturation of protein (denaturation around at normal solution).

Topic: Proteins

Q15. Answer: B — Epithelial cells from damage

Page 422: Keratin is the protein which protects the epithelial cells from damage.

Topic: Proteins

Q16. Answer: B — Glycerol and fatty acid

Page 422: Fats are esters of glycerol and fatty acid. They contain carbon, hydrogen and oxygen but proportionally less oxygen.

Topic: Fats

Q17. Answer: B — 2 – Saturated and Unsaturated fatty acids

Page 422: Only two types – Saturated fatty acids (solid at 20°C, e.g. this is called fat) and Unsaturated fatty acids (liquid form, called oil).

Topic: Fats

Q18. Answer: B — Fat (solid at 20°C temperature)

Page 422: Saturated fatty acids are solid at 20°C temperature, this is called fat.

Topic: Fats

Q19. Answer: B — Fish oil and vegetable oil (liquid form - called oil)

Page 422: Unsaturated fatty acids are found in fish oil and vegetable oil. Examples: coconut oil, palm oil.

Topic: Fats

Q20. Answer: B — 20-30%

Page 422: Normally an adult person should get approximately 20-30% of energy from fat.

Topic: Fats

Q21. Answer: B — 9.3

Page 422: 9.3 kilo calorie energy is liberated from 1 gram of fat.

Topic: Fats

Q22. Answer: B — Gram

Page 422: Fat stores maximum energy per gram.

Topic: Fats

Q23. Answer: B — Prevents the loss of heat and remains under the skin

Page 422: Fat provides energy to the body and prevents the loss of heat, remaining under the skin.

Topic: Fats

Q24. Answer: B — Taste and smell of food

Page 422: Rancidity takes place due to oxidation of fat and oils. It changes the taste and smell of food.

Topic: Fats

Q25. Answer: B — Dry and body weight decreases

Page 422: Due to the lack of fat, skin gets fatty/dry, the body weight of the body gets checked/decreases and development of the body is affected.

Topic: Fats

Q26. Answer: B — Sir F.G. Hopkins

Page 422: Vitamin was coined/invented by Sir F.G. Hopkins.

Topic: Vitamins

Q27. Answer: B — Funk

Page 422: The term vitamin was coined by Funk.

Topic: Vitamins

Q28. Answer: B — 2 – Water soluble and Fat soluble

Page 422: On the basis of solubility, vitamins are of two types: (a) Vitamin soluble in water: Vitamin-B and Vitamin-C, (b) Vitamin soluble in fat: Vitamin-A, Vitamin-D, Vitamin-E and Vitamin-K.

Topic: Vitamins

Q29. Answer: B — B and C

Page 422: Water soluble vitamins are Vitamin B and Vitamin C.

Topic: Vitamins

Q30. Answer: B — A, D, E and K

Page 422: Fat soluble vitamins are Vitamin A, D, E and K.

Topic: Vitamins

Q31. Answer: B — Retinol

Page 422: Vitamin A chemical name is Retinol.

Topic: Vitamins

Q32. Answer: B — Night blindness, Xerophthalmia

Page 422: Vitamin A deficiency causes night blindness and xerophthalmia.

Topic: Vitamins

Q33. Answer: B — Thiamine

Page 422: Vitamin B1 chemical name is Thiamine.

Topic: Vitamins

Q34. Answer: B — Cracking of skin, reddish eye, cracking at corners of lips, mouth ulcers

Page 422: Vitamin B2 deficiency causes cracking of skin, reddish eye, cracking at corners of lips, mouth ulcers.

Topic: Vitamins

Q35. Answer: B — Whitening of hair, mental retardation, Pellagra

Page 422: Vitamin B3 deficiency causes whitening of hair, mental retardation, Pellagra.

Topic: Vitamins

Q36. Answer: B — Burning sensation, Dermatitis

Page 422: Vitamin B5 (Pantothenic acid) deficiency causes burning sensation, dermatitis.

Topic: Vitamins

Q37. Answer: B — Pyridoxine

Page 422: Vitamin B6 chemical name is Pyridoxine.

Topic: Vitamins

Q38. Answer: B — Anemia, skin disease, Depression

Page 422: Vitamin B6 deficiency causes Anemia, skin disease, Depression.

Topic: Vitamins

Q39. Answer: B — Paralysis, body pain, hair falling, scaly and itchy skin

Page 422: Vitamin B7 (Biotin) deficiency causes paralysis, body pain, hair falling, scaly and itchy skin.

Topic: Vitamins

Q40. Answer: B — Folic acid

Page 422: Vitamin B9 chemical name is Folic acid.

Topic: Vitamins

Q41. Answer: B — Pernicious Anemia, jaundice

Page 422: Vitamin B9 deficiency causes Pernicious Anemia, jaundice.

Topic: Vitamins

Q42. Answer: B — Cyanocobalamin

Page 422: Vitamin B12 chemical name is Cyanocobalamin (or Cobalamin).

Topic: Vitamins

Q43. Answer: B — Anemia, dysentery

Page 422: Vitamin B12 deficiency causes Anemia, dysentery.

Topic: Vitamins

Q44. Answer: B — Ascorbic acid

Page 422: Vitamin C chemical name is Ascorbic acid.

Topic: Vitamins

Q45. Answer: B — Scurvy, swelling of gums or bleeding gums

Page 422: Vitamin C deficiency causes Scurvy, swelling of gums or bleeding gums.

Topic: Vitamins

Q46. Answer: B — Milk, Egg, Cheese, Green vegetable, Carrot

Page 422: Vitamin A sources include Milk, Egg, Cheese, Green vegetables, Carrot.

Topic: Vitamins

Q47. Answer: B — Protein deficiency while caloric intake may be adequate

Page 422: Kwashiorkor is caused by protein deficiency, especially when caloric intake is not severely affected.

Topic: Proteins

Q48. Answer: B — Cells, protoplasm and tissues

Page 422: Proteins take part in formation of cells, protoplasm and tissues.

Topic: Proteins

Q49. Answer: B — Pulse, Liver, Egg, Milk, eggs etc.

Page 422: Sources of Vitamin B1 include Pulse, Liver, Egg, Milk, eggs.

Topic: Vitamins

Q50. Answer: B — Beriberi

Page 422: Vitamin B1 (Thiamine) deficiency causes Beriberi.

Topic: Vitamins